GEPM-463

PROJECT MANAGEMENT AND FINANCE

WEEK-08|LECTURE:01

FACILITY LOCATION

WHAT IS FACILITY LOCATION?

In project management, **Facility Location** (also known as plant location) is the strategic process of determining the optimal geographical site for a company's physical operations, such as a software development center, a data center, or an office.

This decision is critical because it involves **long-term commitment and substantial capital investment** that cannot be easily reversed. A well-chosen location contributes to operational efficiency and profitability by ensuring access to customers, skilled labor, and necessary infrastructure.

KEY ASPECTS OF FACILITY LOCATION

Selecting a facility location involves three distinct levels of analysis:

A. **Selection of a Region:** Choosing a broad geographical area (e.g., a specific country or a state like Dhaka or Rajshahi).
B. **Selection of a Locality:** Identifying a specific community or district within that region.
C. **Selection of a Site:** Picking the exact plot of land or building.

WHY THESE ASPECTS ARE CRITICAL?

- **Selection of a Region:** (Selecting between Dhaka Division or Rajshahi Division)
  - **Proximity to Talent:** Does the region have top CS universities? (e.g., proximity to BUET/MIST/DU in Dhaka).
  - **Internet Infrastructure:** Availability of good ISP gateways and fiber-optic backbones.
  - **Suitability of Climate:** Cold weather reduces cooling costs for server.

WHY THESE ASPECTS ARE CRITICAL?

- **Selection of a Locality:** (Choosing between Gulshan or Kawran Bazar)

  - **Power Reliability:** Is the area on a "preferred grid" with minimal load-shedding?
  - **Communication Facilities:** High-speed internet and reliable & fast communication facilities.
  - **Local Taxes & Laws:** Incentives offered by the Bangladesh Hi-Tech Park Authority (BHTPA) such as tax holidays for 10 years.
  - **Living Conditions:** Is it easy for software engineers to commute or live nearby?

WHY THESE ASPECTS ARE CRITICAL?

- **Selection of a Site:** (Choosing a specific plot/room/floor)
  - **Price of Land/Lease:** Balancing the high cost of premium urban office space against the lower cost of suburban data center plots.
  - **Disposal of Waste (E-Waste):** IT sites must have protocols for the safe disposal of toxic electronic components, old servers, and lithium batteries.
  - **Availability for Expansion:** For a growing startup, the site must have "adequate land availability" to build additional server halls or larger "Agile pods" as the team scales.

LOCATION BREAK-EVEN ANALYSIS

The economic comparison of location alternative is facilitated by the use of cost volume profit analysis known as location break-even analysis.

The analysis can be done numerically/graphically. Following steps are involved in the break-even analysis:

a. Determine the fixed and variable costs associated with each location alternative.
b. Calculate the total cost for all locational alternatives.
c. Determine the location having the lowest total cost for the expected level of output.

LOCATION BREAK-EVEN ANALYSIS

Imagine a Dhaka-based IT firm is deciding where to host its new SaaS platform for **2,000 monthly active users.**

Location Option	Fixed Cost (Servers, Rent, Licenses)	Variable Cost (Bandwidth, Electricity, Support)
Site A: Kawran Bazar Office	Tk. 50,000	Tk. 80 / user
Site B: Kaliakair Hi-Tech Park	Tk. 80,000	Tk. 40 / user
Site C: Cloud (AWS/Azure)	Tk. 20,000	Tk. 120 / user

Total Cost Calculation= (TC = FC + VC *Volume)

LOCATION BREAK-EVEN ANALYSIS

Site A: 50,000 + (80 *2,000) = 210,000 BDT

Site B: 80,000 + (40*2,000) = 160,000 BDT (Most Economical)

Site C: 20,000 + (120 *2,000) = 260,000 BDT

Lesson: For high-volume projects, the Hi-Tech Park (Site B) becomes more profitable despite the higher upfront setup costs.

LOCATION BREAK-EVEN ANALYSIS

A, B and C are the three potential locations for producing “X” expected to sell for BDT 90. Find the most economical location for an expected volume of 1,850 units/year.

Location Option	Fixed Cost	Variable Cost
Site A	Tk. 20,000	Tk. 50 / user
Site B	Tk. 40,000	Tk. 30 / user
Site C	Tk. 80,000	Tk. 10 / user

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PROJECT MANAGEMENT AND FINANCE

WEEK-08|LECTURE:02

PRACTICE PROBLEMS: LOCATION BREAK-EVEN ANALYSIS

A software company is considering three locations to open a new development office.

Location Option	Fixed Cost (Tk./year)	Variable Cost (Tk./employee/year)
Site A: Kawran Bazar Office	Tk. 600,000	Tk. 25,000
Site B: Kaliakair Hi-Tech Park	Tk. 900,000	Tk. 18,000
Site C: Cloud (AWS/Azure)	Tk. 1,200,000	Tk. 12,000

Task01: Determine the most economical location for:

(a) 20 employees (b) 50 employees (c) 100 employees

Task02: Draw the cost volume graph.

PRACTICE PROBLEMS: LOCATION BREAK-EVEN ANALYSIS

A firm producing IoT devices is choosing a plant location.

Location Option	Fixed Cost (Tk./year)	Variable Cost (Tk./Unit)
Plant A	Tk. 500,000	Tk. 70
Plant B	Tk. 800,000	Tk. 50
Plant C	Tk. 120,000	Tk. 30

Task01: Calculate total cost for an annual output of 10,000 units.

Task02: Determine the best location for: (a) Low volume (≤ 5,000 units)(b) Medium volume (≈ 10,000 units)(c) High volume (≥ 20,000 units).

PRACTICE PROBLEMS: LOCATION BREAK-EVEN ANALYSIS

A software company wants to establish a new development center. The expected annual workload is measured in number of projects completed per year.

Location Option	Fixed Cost (Tk./year)	Variable Cost (Tk./Project)
A	Tk. 500,000	Tk. 18,000
B	Tk. 420,000	Tk. 20,000
C	Tk. 300,000	Tk. 24,000
D	Tk. 320,000	Tk. 22,000
E	Tk. 280,000	Tk. 23,000

Task01: Determine the most economical location if the expected workload is 40 projects/year. **Task02:** Draw the cost volume graph.

Thank You

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PROJECT MANAGEMENT AND FINANCE

WEEK-09|LECTURE:01

MAINTENANCE
PLANNING AND MANAGEMENT

WHAT IS MAINTENANCE?

Maintenance is the process of **keeping a system, product, software, or asset in proper working condition** after it has been developed or installed.

It includes:

- Fixing problems
- Improving performance
- Updating or replacing parts/systems
- Preventing future failures

Maintenance ensures reliability, efficiency, safety, and long-term usability.

MAINTENANCE TAXONOMY
(Based on Application Domain)

In Software Engineering:
We call it Software Maintenance

In Engineering / Project / Asset Management:
We call it Asset Maintenance or Industrial Maintenance

SOFTWARE MAINTENANCE

Software Maintenance is the process of modifying, updating, and improving software after it has been delivered, to ensure it continues to function correctly and efficiently.

Software maintenance is required to:

- Fix bugs and errors
- Adapt to new operating systems or hardware
- Improve performance
- Add new features
- Enhance security
- Minimize unplanned downtime and avoid costly disruptions.
- Ensure ongoing compliance and reduce regulatory risks

WHAT ARE THE TYPES OF SOFTWARE MAINTENANCE?

Corrective maintenance

Bug fixes, error correction, patching vulnerabilities

Adaptive maintenance

Dependency updates, configuration changes, compliance updates

Perfective maintenance

Feature enhancements, UI improvements, code cleanup

Preventive maintenance

Refactoring, removing obsolete code, documentation updates

Source: N-iX

CORRECTIVE MAINTENANCE

This type of maintenance focuses on addressing bugs, logic errors, and system faults identified after deployment.

Key activities: Debugging, deploying patches, and applying quick fixes to repair defects and restore functionality.
Trigger: Usually triggered by user reports, QA feedback, or system monitoring alerts.
Example: Resolving a login failure that occurs following the release of a new software version.

ADAPTIVE MAINTENANCE

Adaptive maintenance ensures that your software remains compatible with changes in its operating environment, such as new OS versions, third-party APIs, or updated regulations.

Key activities: Upgrading dependencies, adjusting configurations, and implementing compliance-related updates.
Trigger: Triggered by changes in platforms, technology stacks, or regulatory standards
Example: Updating cloud infrastructure integration of a specific AWS service.

PERFECTIVE MAINTENANCE

The key goal of perfective maintenance is to enhance software functionality, performance, and user experience based on evolving business needs and user feedback.

Key activities: Adding new features, optimizing algorithms, improving user interfaces, and cleaning up code for better readability and maintainability.
Trigger: Typically driven by product growth, insights from user behavior, or strategic roadmap planning.
Example: improving page load speed or redesigning a dashboard layout to enhance usability and meet updated user expectations.

PREVENTIVE MAINTENANCE

Preventive maintenance is a forward-looking approach that focuses on identifying and addressing potential issues before they impact system performance.

Key activities: Code refactoring, removing risky components, updating documentation, and improving architectural consistency.
Trigger: Typically initiated after code reviews, technical audits, or architectural assessments that reveal potential risks.
Example: Rewriting legacy modules to simplify future development work.

PRACTICE PROBLEMS

SCENARIO 1: A hospital management system is producing incorrect patient bills due to a calculation error discovered after deployment. What type of software maintenance is required?
Answer: Corrective Maintenance.

SCENARIO 2: An organization upgrades its operating system, and the existing payroll software no longer runs properly.
Answer: Adaptive Maintenance.

SCENARIO 3: Developers restructure messy and poorly written code to improve readability and reduce future bugs, even though no error currently exists
Answer: Preventive Maintenance.

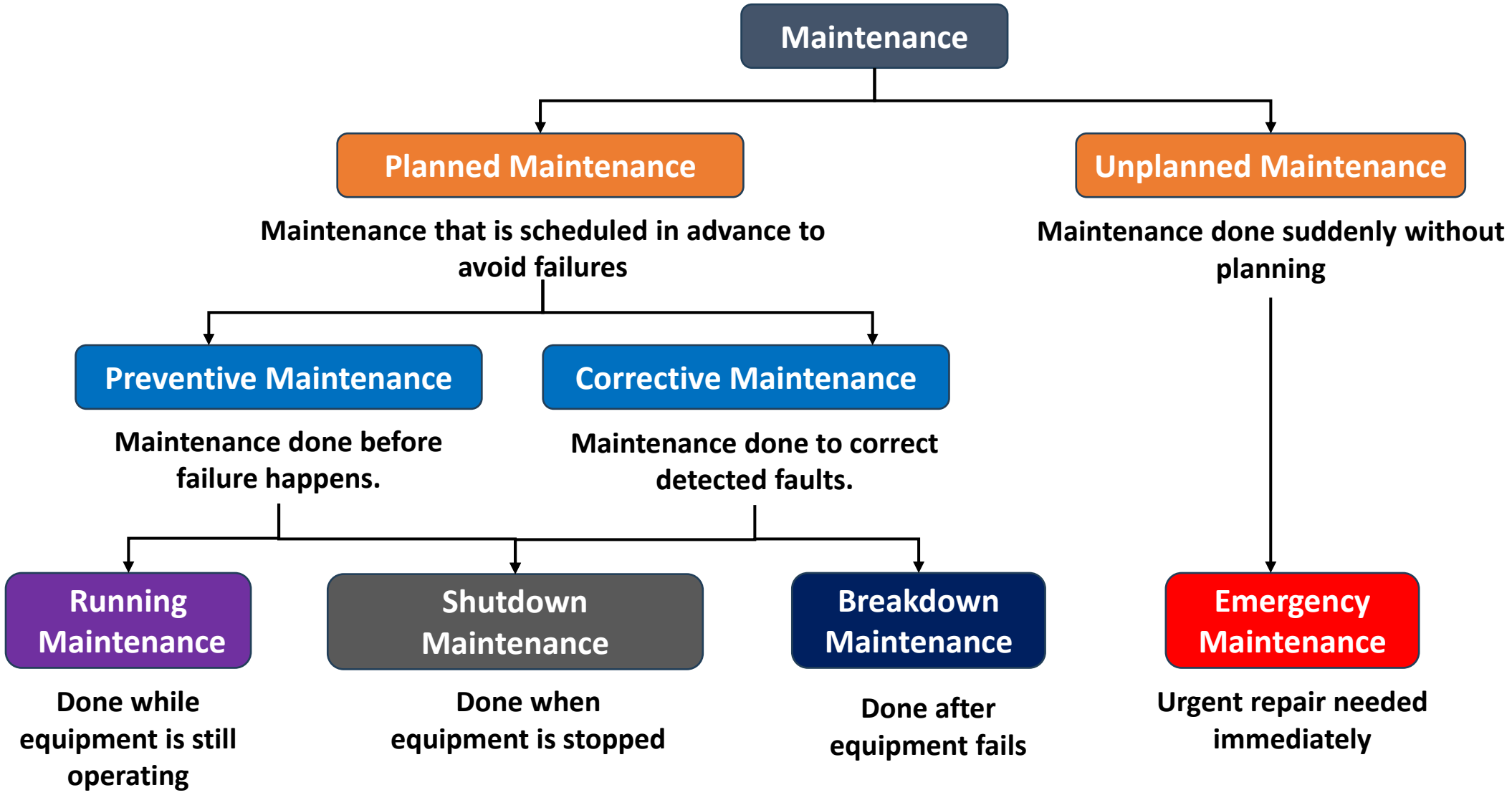
ASSET MAINTENANCE OR INDUSTRIAL MAINTENANCE

Asset Maintenance—also called **Industrial Maintenance**—refers to the systematic process of maintaining, repairing, and optimizing physical assets (machines, equipment, infrastructure, tools, and facilities) to ensure they operate efficiently, safely, and reliably.

Objectives of Asset Maintenance:

1. To maximize the availability and reliability of all assets.
2. To obtain the maximum possible return on investment.
3. To extend the life of assets by minimizing wear and tear and deterioration.
4. To ensure operational readiness for emergency (standby units, fire-fighting and rescue units, etc.)
5. To ensure the safety of personnel using facilities.

TYPES OF MAINTENANCE (ASSET)



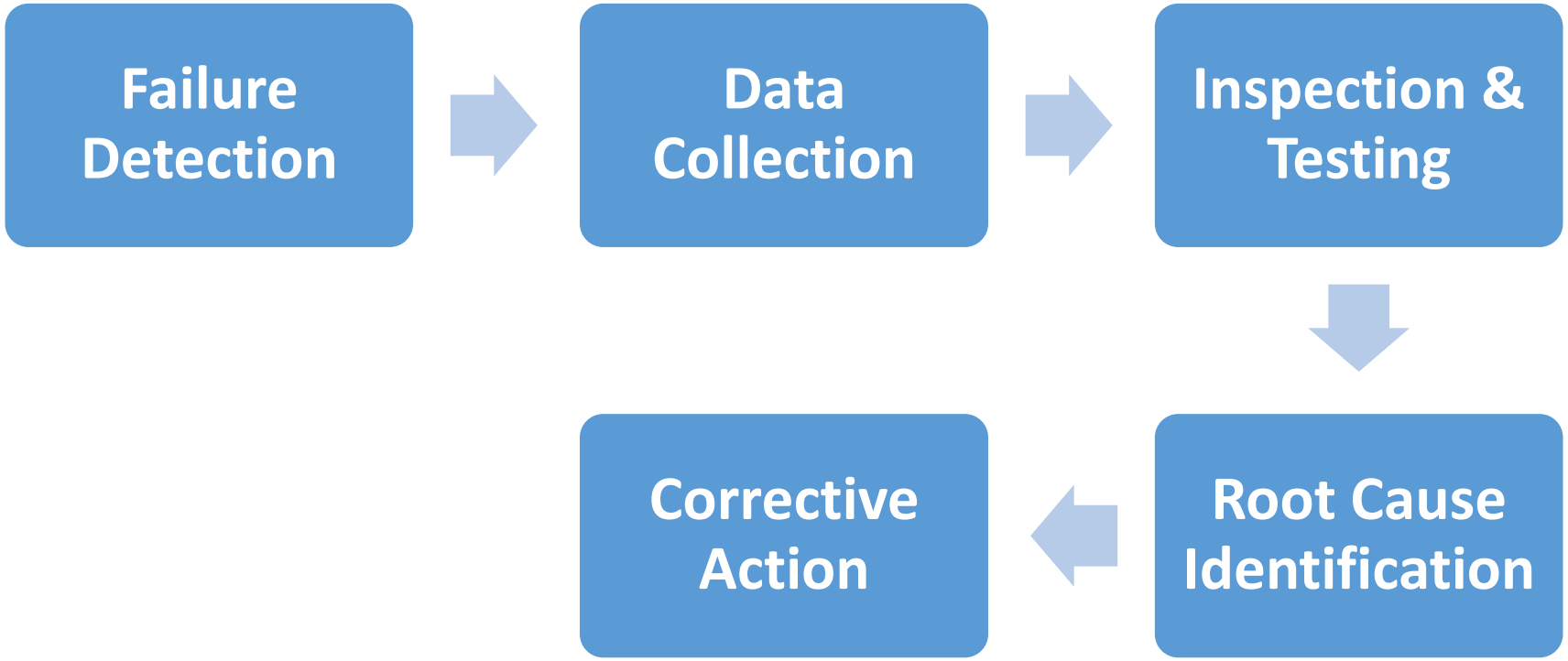
FAILURE ANALYSIS

Failure Analysis is a **systematic investigation process** used to determine **why** a component, system, or equipment failed, **how** the failure happened, and **how** to prevent it from happening again.

Main Goals of Failure Analysis:

- Identify the root cause of failure.
- Prevent repeated failures.
- Improve product or system design.
- Improve maintenance strategies.
- Reduce operational cost and downtime.
- Improve safety and reliability.

STEPS IN FAILURE ANALYSIS PROCESS



BATH-TUB CURVE

The Bath-Tub Curve is a concept in reliability engineering that shows how the failure rate of a product or equipment changes over its lifetime. It plays a vital role in taking decisions pertaining to maintenance planning and control.

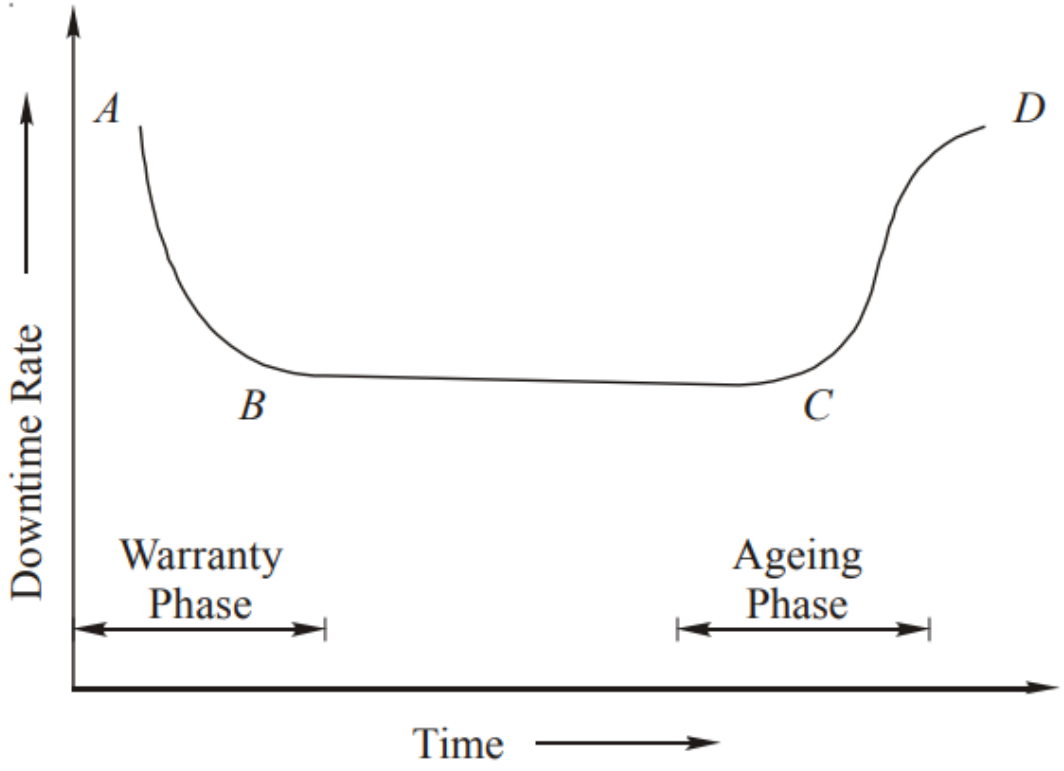
It is called a bath-tub curve because its shape looks like a bathtub:

- **High failure rate** at the **beginning**
- **Low, stable failure rate** in the **middle**
- **Increasing failure rate** at the end

BATH-TUB CURVE

Why It Is Important?

- Helps plan maintenance strategy
- Supports preventive & predictive maintenance
- Helps decide warranty periods
- Helps estimate product life
- Reduces unexpected breakdowns



Three Stages of the Bath-Tub Curve

Stage 1: Early Failure Period (Infant Mortality):

- Failure rate is high at the beginning
- Failures decrease over time

Stage 2: Useful Life Period (Normal Operation)

- Failure rate is low and almost constant
- Equipment works reliably

Stage 3: Wear-Out Period (End of Life)

- Failure rate increases over time
- Components wear out

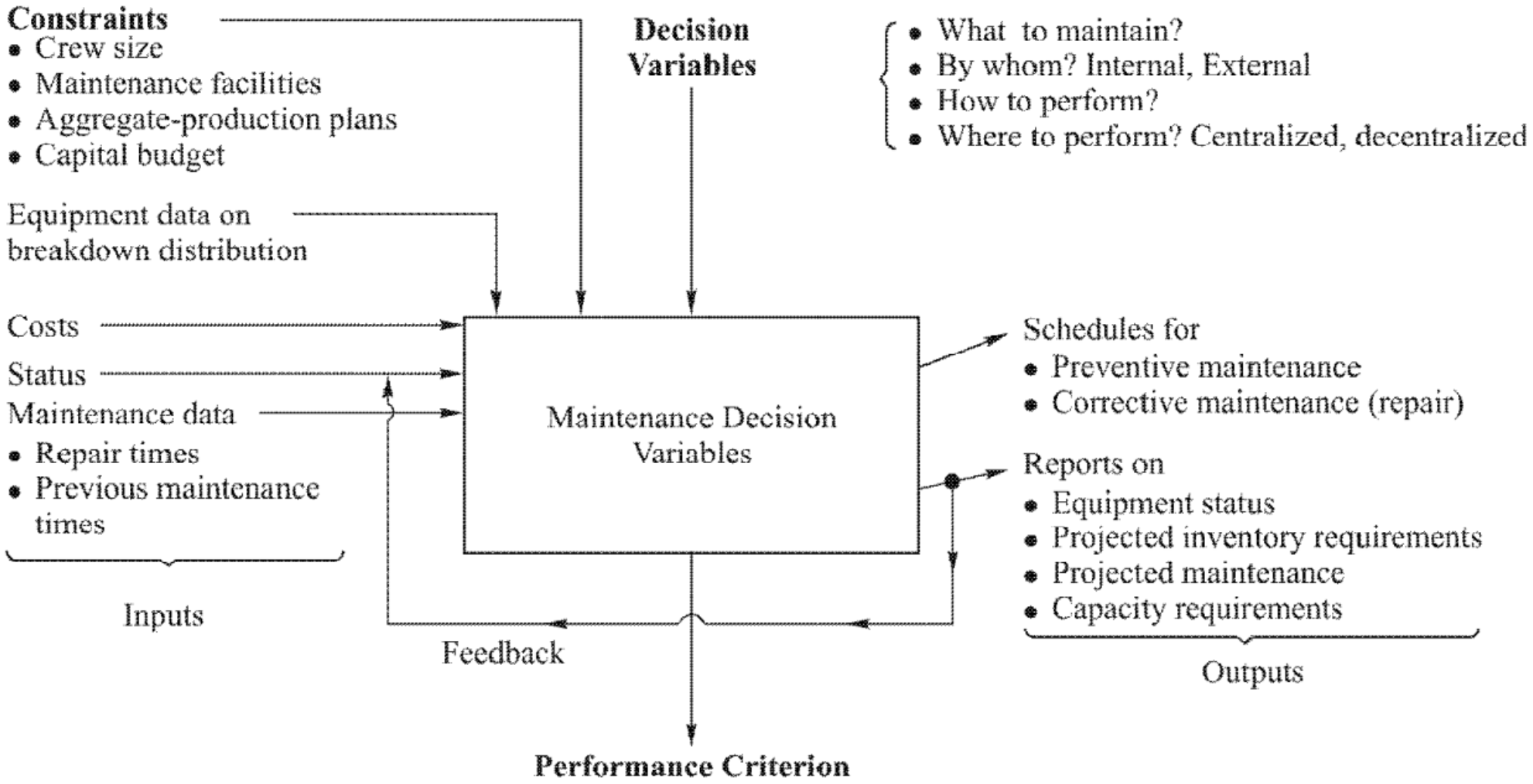
OPERATION MANAGEMENT MAINTENANCE SYSTEM

An Operations Management Maintenance System is a structured approach used by organizations to plan, schedule, control, and optimize maintenance activities so that equipment, machines, and facilities operate efficiently, reliably, and cost-effectively.

It is a system that helps organizations decide:

- What maintenance to do
- When to do maintenance
- How to do maintenance
- Who will do maintenance
- How much it will cost

BASIC COMPONENTS OF AN OPERATION MANAGEMENT MAINTENANCE SYSTEM



BASIC COMPONENTS OF AN OPERATION MANAGEMENT MAINTENANCE SYSTEM

1. **Constraints (Top Left)**

These are limitations within which maintenance decisions must be made:

- Crew size (available technicians),
- Maintenance
- Capital budget, etc.

2. **Inputs (Left Side – Data for Decision Making)**

These are the information sources used to make decisions:

Equipment Data: failure patterns

Costs: Maintenance costs, repair costs

Status Information: Current equipment condition, Maintenance history, Repair times, Previous maintenance records

3. Maintenance Decision Variables (Center Box):

This is basically choosing the maintenance strategy.

- What to maintain? By whom? (Internal staff or external contractor)
- How to perform maintenance? Where to perform?

4. Outputs (Right Side):

After decisions are made, the system produces:

Schedules:

Preventive Maintenance

Corrective Maintenance (repair)

Reports

- Equipment status
- Projected inventory requirements (spare parts)
- Projected maintenance workload
- Capacity requirements

BASIC COMPONENTS OF AN OPERATION MANAGEMENT MAINTENANCE SYSTEM

5. Feedback Loop (Bottom Arrow)
After implementation:
- Results are monitored
- Performance is measured
- Costs are evaluated

This feedback goes back into the system to improve future decisions.

6. Performance Criterion (Bottom)
Total Maintenance Cost =
- Cost of preventive maintenance
- Cost of corrective maintenance
- Cost of downtime (production loss).

THANK YOU

1. What is Kano Model?

When developing a product, service, or software, it's impossible to include every feature immediately, as many might demand. Instead, it's crucial to determine what to prioritize, what to schedule for later, and what to exclude altogether. Making these decisions effectively requires the support of data. So, we need to create a product roadmap with the right features.

To make this process easier, Noriaki Kano, a Japanese researcher and consultant, published a paper in 1984 with a set of ideas and techniques which is commonly known as **Kano Model**. The **Kano Model** is a **framework** for understanding and prioritizing **customer needs and satisfaction** in **product development and service design**. It categorizes **features or requirements** based on how they **influence customer satisfaction**.

2. What Are those Categories?

A. **Basic (Must-be):** Essential features that customers take for granted. If absent, customers are dissatisfied, but their presence does not significantly increase satisfaction.
B. **Performance (One-dimensional):** Features where customer satisfaction is directly proportional to the performance level. The better the feature, the happier the customer.
C. **Delighters (Attractive Needs):** Features that customers do not expect but are pleasantly surprised by when present. Their absence does not cause dissatisfaction.
D. **Indifferent Needs:** Features that neither improve nor harm customer satisfaction, regardless of their presence or absence.
E. **Reverse Needs:** Features that satisfy some customers but annoy others. These are polarizing and require careful analysis of the target audience.

3. How We Should Collect Data for Kano Analysis?

We can use just survey questionnaire to collect data for Kano Analysis. To do so, potential participants are asked two questions about each potential feature:

Functional form: How do you feel if the feature is present?

Dysfunctional form: How do you feel if the feature is absent?

Responses are typically collected using predefined options such as:

(1) I like it: The respondent is enthusiastic about the feature and sees it as highly desirable.

(2) I expect it: The respondent views the feature as a basic requirement or must-have. They assume it will be included and take it for granted.

(3) I am neutral: The respondent has no strong feelings about the feature—neither positive nor negative.

(4) I can tolerate it: The respondent is not happy with the absence or poor performance of the feature, but they can accept it without becoming completely dissatisfied.

(5) I dislike it: The respondent strongly opposes the feature or its absence. Meaning, If the feature is present, it causes dissatisfaction.

Example (Survey): **Functional form:**

How do you feel if the feature is present?

Feature 1	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 2	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 3	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 4…	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 10	I like it	I expect it	I am neutral	I can tolerate it	I dislike it

How do you feel if the feature is absent? **Dysfunctional form**

Feature 1	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 2	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 3	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 4…	I like it	I expect it	I am neutral	I can tolerate it	I dislike it
Feature 10	I like it	I expect it	I am neutral	I can tolerate it	I dislike it

$$SatisfactionCoefficient = \frac{A + O}{A + O + M + I}$$

$$DisatisfactionCoefficient = \frac{O + M}{A + O + M + I}$$

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PROJECT MANAGEMENT & FINANCE

WEEK-13

NETWORK DIAGRAM IN PROJECT MANAGEMENT

First, share your thoughts

WHAT IS A PROJECT ?

WHAT IS A PROJECT?

A project is a SET OF TASKS (TEMPORARY) done to achieve a specific goal within a limited time

Project = Temporary + Unique + Goal

LET'S IMAGINE A SCENARIO

MIST is Organizing a University Tech Fest

MIST is planning to organize a Tech Fest next month, and a group of students from CSE Dept. has been assigned to manage the entire event within a limited timeframe. Initially, the team needs 2 days to decide the event theme and finalize the date. After that, they require 3 days to prepare a budget and 2 days to get approval from the university authority. Once approval is granted, the team proceeds to book the venue (2 days) and arrange necessary logistics such as chairs, stage, and sound system, which takes 4 days. At the same time, the marketing team starts designing posters (3 days) and then continues with promotion on social media for 5 days. Meanwhile, another group works on inviting guest speakers (3 days) and confirming their participation (2 days). As promotion progresses, participant registration opens and continues for 4 days. Finally, after all preparations are completed, the Tech Fest event is executed in 1 day.

Identify all the project activities from the above scenario & arrange them in sequence….

ACTIVITIES IN SEQUENCE WITH DURATION

Activity	Description	Time
A	Decide theme & date	2 days
B	Prepare budget	3 days
C	Get approval	2 days
D	Book venue	2 days
E	Arrange logistics	4 days
F	Design posters	3 days
G	Promotion	5 days
H	Invite speakers	3 days
I	Confirm speakers	2 days
J	Registration	4 days
K	Event execution	1 day

ACTIVITY DEPENDENCY TABLE

ID	Activity	Duration (Days)	Predecessor(s)	Explanation
A	Decide theme & date	2	—	First task, no dependency
B	Prepare budget	3	A	Budget can start only after theme is decided
C	Get approval	2	B	Approval depends on budget
D	Book venue	2	C	Venue booking needs approval
E	Arrange logistics	4	D	Logistics after venue booking
F	Design posters	3	C	Poster design starts after approval
G	Promotion	5	F	Promotion after posters are ready
H	Invite speakers	3	C	Invitations sent after approval
I	Confirm speakers	2	H	Confirmation after invitation
J	Registration	4	G	Registration opens after promotion starts
K	Event execution	1	E, I, J	Final event after all preparations

NETWORK DIAGRAM

Solved in the Classroom

CRITICAL PATH IDENTIFICATION

Solved in the Classroom